

The empirical recurrence for OEIS A233729: recovering a conjecture that is not written down

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Abstract

OEIS A233729 records “Empirical recurrence of order 84”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 1024$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 84, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 84 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A233729 is “Number of $(n+1) \times (4+1)$ 0..3 arrays with every 2×2 subblock having the sum of the absolute values of the edge differences equal to 6 and no adjacent elements equal.”. It has offset 1 and begins

2104, 21432, 233432, 2723072, 32240288, 389230764, 4696134192, 57174669348, ...

The entry states:

Empirical recurrence of order 84 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 08:10:31, revision 8) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1024$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 1024$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 2048$ exact terms of the model returns order 84 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{84} c_i a(n-i),$$

c_1	7	c_{22}	10561358420436036	c_{43}	−69247747475084737346367	c_{64}	184358655
c_2	331	c_{23}	−444828379696067297	c_{44}	55814220061985012389187	c_{65}	82062983
c_3	−2299	c_{24}	9388670429410821	c_{45}	91578642055587430699421	c_{66}	−31996000
c_4	−43599	c_{25}	3193690667667779718	c_{46}	−73982918068171036622250	c_{67}	−13066964
c_5	308388	c_{26}	−644783192089736124	c_{47}	−102506715264583092375426	c_{68}	42444794
c_6	3225045	c_{27}	−19120065458077633225	c_{48}	82027750512223906216224	c_{69}	16046619
c_7	−23626157	c_{28}	6636058808909198928	c_{49}	96913181267583949506557	c_{70}	−4191146
c_8	−154357300	c_{29}	95942457860070592102	c_{50}	−75921073231649690863811	c_{71}	−1482273
c_9	1189570775	c_{30}	−44525876389594748782	c_{51}	−77175159080793281642600	c_{72}	2972670
c_{10}	5154033359	c_{31}	−405138787530032794224	c_{52}	58476912331679578275460	c_{73}	9970260
c_{11}	−42517012973	c_{32}	226199326901877080932	c_{53}	51579066410668010037846	c_{74}	−144020
c_{12}	−125481638475	c_{33}	1444144157409722028589	c_{54}	−37323404707415855648501	c_{75}	−46751
c_{13}	1132197280604	c_{34}	−915182316009836813330	c_{55}	−28800228509217256057208	c_{76}	441377
c_{14}	2285052378114	c_{35}	−4355370777020357434834	c_{56}	19632356559372915300310	c_{77}	143818
c_{15}	−23214586963293	c_{36}	3018992970192733942178	c_{57}	13359366825295108384144	c_{78}	−7460
c_{16}	−31452752653342	c_{37}	11130998152417867836951	c_{58}	−8452289320924683399028	c_{79}	−2657
c_{17}	375258379553901	c_{38}	−8223423699006245981763	c_{59}	−5112225105112669652457	c_{80}	473
c_{18}	324713660314975	c_{39}	−24129304731339648011690	c_{60}	2953231202898007647896	c_{81}	2570
c_{19}	−4866962594699561	c_{40}	18630748450209644330167	c_{61}	1600104513902578288220	c_{82}	230
c_{20}	−2403148718885480	c_{41}	44381675932622956040099	c_{62}	−828701214001558128960	c_{83}	−9
c_{21}	51329707609438012	c_{42}	−35251118450077269092840	c_{63}	−405369625833437960382	c_{84}	−2

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 84, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $84 + 4$ terms removed returns order 84 as well, so $\deg q_{\min} = 84 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $84 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 84 that this sequence satisfies — and that family turns out to have exactly one member.

References

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